

Evaluating Linear and Non-Linear Classification Methods Using Synthetic Health Data

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and a major contributor to cognitive decline worldwide. Early identification of elevated AD risk may improve intervention and management strategies. This project applied multiple predictive modeling approaches to a publicly available synthetic Alzheimer's disease dataset ($n = 2,149$; 33 variables) to classify individuals at elevated risk of AD. Methods included logistic regression, best subset logistic regression, lasso regularization, k-nearest neighbors (kNN), and gradient boosting machine (GBM). Model performance was evaluated using repeated 5-fold cross-validation over 300 iterations. For each iteration, mean fold-specific performance metrics were recorded and used to estimate overall model performance and confidence intervals.

Section 1: Data and Preparation and Exploratory Data Analysis

Goal and Data Set

The primary objective of this project was to develop and compare classification models for identifying individuals at elevated risk of Alzheimer's disease. utilized a publicly available synthetic dataset from Kaggle containing 2,149 observations and 33 predictors across different categories of health-related measures, including demographic, lifestyle, clinical, and functional health domains. The full list of variables and their meanings can be found from Table 1.1 in the appendix.

Evaluation Metrics & Methods

We employed a series of performance metrics to assess each classifier's performance on identifying people with Alzheimer's disease in the data set. The metrics include Recall (proportion of actual positive cases the classifier can correctly capture), Precision (proportion of predicted positive cases that are actually positive), accuracy (proportion of correct predictions), and AUC (classifier's overall ability of classifying positive and negative cases). We prioritized the recall in the training of classifier, as certain level of false positive result is acceptable and inevitable due to the aim of early-stage screening tools, which is exhaustively detect potential positive cases and refer them to more accurate, but complicated high-cost diagnosis (White & Algeri, 2023). Model performance was evaluated using repeated 5-fold cross-validation over 300 iterations. For each iteration, mean fold-specific performance metrics were recorded and used to estimate overall model performance and confidence intervals.

A table of variables used by each classifier can be found in appendix(Table 1.6).

Exploratory Data Analysis

*Tables and figures can be found in the appendix by referring to table/figure numbers.

We explored the distribution of all observations for demographic variables and the relationship between certain variables concerning disease status. There were 1389 patients with positive diagnosis results and 760 with negative diagnosis results, where the data set is not severely imbalanced (Figure 1.1). No imbalanced gender distribution was found in the data set (Figure 1.2).

Among people with a positive diagnosis of AD, 13.6% of people had diabetes (see appendix Table 1.2), 19.7% had depression (Table 1.3), 28.6% of the people with AD were cigarette users (Table 1.4), 23.3% had family history of Alzheimer's (Table 1.5). There were 22.9% who did not receive any education, 39.7% with high school degrees, 28.6% of them held bachelor's degrees and 8.8% of them holding higher degrees. People with high school degrees as highest degree were the largest group in the population with positive AD diagnosis (Figure 1.3). Our scatter plot of Age vs MMSE respect do AD status indicated that people with higher MMSE (Mini Mental State Exam) score tend to be less likely to suffer from AD, and this pattern existed across the span of age (Figure 1.4). A set of 4 cholesterol-related variables were biologically correlated, thus, only 1 of the 4 variables that have strong association with the AD was retained (Martinez et al., 2022, CDC, 2024). No missing values were found in the data set (Figure 1.5).

Section 2: Linear Models

MLR

We began by fitting a multiple logistic regression model using all the available variables in the data set. A repeated 5-fold Cross Validation with 300 iterations was used to validate the performance of this classifier.

Table 1. Performance Metrics of MLR Model with All Available Variables

Metrics	Performances
Recall	0.728
Recall CI	(0.7273,0.7285)
Precision	0.793
Precision CI	(0.7919, 0.7931)
Accuracy	0.836
Accuracy CI	(0.8357, 0.8364)
AUC	0.8979
AUC CI	(0.8977, 0.8980)

According to the outcome from repeated 5-fold Cross Validation, the multiple logistic regression classifier The full logistic regression model achieved a recall of 0.728, precision of 0.793, accuracy of 0.836, and AUC of 0.898, indicating solid baseline discriminatory performance while preserving interpretability.(Prager, 2023). However, including all the available variables in the model could induce the issue of overfitting the model, therefore worsening the classifier's performance in a real-world setting, other methods shall be utilized to address this issue.

MLR with Best Subset

We first attempted to integrate the best subset selection technique with the MLR model to create the second classifier. The best subset selection was achieved through regsubset() function in R. The following variables were used in the MLR model with best subset: Age, Sleep, CholesterolHDL, MMSE, FunctionalAssessment, MemoryComplaints, Behavior Problems.

Table 2. Performance Metrics of MLR Classifier with Best Subset Selection

Metrics	MLR Best Subset
Recall	0.742
Recall CI	(0.7416,0.7424)
Precision	0.8044
Precision CI	(0.8040, 0.8047)
Accuracy	0.8446
Accuracy CI	(0.8440, 0.8447)
AUC	0.903
AUC CI	(0.90288, 0.90304)

The MLR with best subset variables allowed us to reach a recall of 0.742(95%CI: 0.7417, 0.7424), a precision of 0.804 (95% CI: 0.8040, 0.8047), an accuracy of 0.845 (95% CI: 0.8444,0.8448), and an AUC of 0.903 (95% CI: 0.90288,0.90304). A recall of 0.742 indicates that the classifier correctly detect 74.2% of the positive cases of Alzheimer's disease in all patients. The precision of 0. 804 suggested that 80.4% of the predicted positive cases were true positive AD cases. An accuracy of 0. 845 indicated that the MLR classifier along with best subset of variables was able to correctly classify 84.5% of the cases. An AUC of 0. 903 suggested that the classifier's discriminatory performance between two classes was good.

The performance metrics suggested that the classifier build through MLR with best subset selection technique was able to correctly predict for 74.2% of people on their positive AD status. 80.44% of the predicted positive cases were true positive cases. 84.46% of the cases were correctly classified, where as an AUC of 0.903 indicated a good ability for the classifier to distinguish between two classes.It can be observed that the utilization of Best subset selection produced only modest performance gains relative to the full logistic regression model, suggesting limited incremental predictive benefit despite reduced feature dimensionality.

Lasso

Lasso Regularization, which was a regularization technique aimed at resolving the issue of overfitting and performing variable selection simultaneously. It also allows the coefficients of certain correlated variables to be shrink to zero, therefore avoiding the impacts of multicollinearity. The mathematical expression of lasso shown as follows, our goal is to find the $\hat{\beta}$ that minimizes the following:

$$\hat{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^n \left(y_i - \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)}} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right)$$

In order to implement the lasso regularization for the MLR model, we need to find the value of tuning parameter λ that provides the most identical shrinkage effect to the model(i.e. provide the lowest model deviance). To find the best value of λ , we utilized 10-fold Cross Validation along with a series of values from 10^{10} to 10^{-2} on a logarithmic scale. (See Figure 2.1 in appendix). Lowest deviance occurred when $\lambda = 0.00612559$. Therefore, we select this value of λ and fit the MLR model with lasso regularization.

Lowest deviance occurred when $\lambda = 0.00612559$. Therefore, we select this value of and fit the MLR model with lasso regularization and utilized repeated cross validation to assess the performance of the model. The following variables were used in the final model of Lasso: Age, EducationLevel, Smoking, SleepQuality, CardiovascularDisease, HeadInjury, Hypertension, CholesterolHDL, MMSE, FunctionalAssessment, MemoryComplaints, BehavioralProblems, ADL, Confusion, Disorientation.

Table 3. Performance Metrics of MLR Classifier with Lasso Regularization

Metrics	Lasso
Recall	0.7286
Recall CI	(0.7281,0.7290)
Precision	0.8083

Metrics	Lasso
Precision CI	(0.8078, 0.8088)
Accuracy	0.8426
Accuracy CI	(0.8423, 0.8428)
AUC	0.9011
AUC CI	(0.90105, 0.90125)

The MLR with lasso regularization allowed us to reach a recall of 0.7286 (95%CI: 0.7281, 0.7290), a precision of 0.8083 (95% CI: 0.8078, 0.8088), an accuracy of 0.8426 (95% CI: 0.8423, 0.8428), and an AUC of 0.9011 (95% CI: 0.90105, 0.90125). The results indicated the MLR classifier with lasso regularization correctly identified 72.86% of the positive cases among all actual positives and 80.83% of its predicted positive cases were true positives. An accuracy of 0.8426 suggested that the classifier was able to correctly classify 84.26% of all cases into their respective classes. The AUC of 0.9011 shows an excellent ability to differentiate between positive and negative cases. These results demonstrated that Lasso regularization improved feature selection efficiency and reduced model complexity but did not substantially outperform alternative linear modeling approaches in predictive performance.

Section 3. Non-Linear Models

kNN

K-Nearest Neighbors (kNN) classifies data points based on the value of k nearest points in the variable space. It is easy to implement on data due to the absence of an explicit training period but does poorly with large datasets and noisy data due to its sensitivity to irrelevant features. This algorithm was employed as an alternative to the multiple logistic regression model classifier on our dataset. Repeated 5-fold cross-validation with 300 iterations was also applied to evaluate the performance of this classifier. The Recursive Feature Elimination (RFE) method was leveraged to address kNN's issue on dealing with data consists of too many variables. RFE is a method that repetitively fitting the model with data provided and remove the least important variable for prediction. A range of different functions, including linear, random forest, etc. is available in R as the function for RFE to use to perform variable selection. We chose random forest as the function to select a subset of important variables for kNN to use to construct the classifier. The following variables were selected by RFE for the building of kNN classifier: FunctionalAssessment, ADL, MMSE, DietQuality, SleepQuality.

We utilized 10-fold Cross Validation along with a series of different values of k from 1 to 20 to find the best k for kNN to use as number of nearest data points during prediction. The best k was found to be 1. We utilized repeated 5-fold Cross Validation with 300 iterations to validate the performance of kNN classifier.

Table 4. Performance Metrics of kNN classifier

Metrics	kNN
Recall	0.637
Recall CI	(0.6368,0.6387)
Precision	0.642
Precision CI	(0.6411, 0.6429)
Accuracy	0.7458
Accuracy CI	(0.7452, 0.7463)
AUC	0.721
AUC CI	(0.7207, 0.7218)

According to the results, the kNN classifier achieved a recall of 0.638 (95% CI: 0.6368, 0.6387), a precision of 0.642 (95% CI: 0.6412, 0.6429), an accuracy of 0.746 (95% CI: 0.7452, 0.7463), and an AUC of 0.721 (95% CI: 0.7207, 0.7219). In other words, this classifier was able to detect 63.8% of the positive AD cases in all patients with positive AD diagnosis status, and 64.2% of the predicted positive cases were true positives. An accuracy of 74.6% suggested that the classifier was able to correctly classify 74.6% of the patients into correct classes. The AUC of 0.721 indicated a fair rating of classifier's discriminatory performance between positive and negative cases of AD. While the kNN classifier demonstrated moderate performance across all metrics, it underperformed compared to other classifier's, particularly in recall and AUC. This indicates that kNN struggled to effectively detect Although the optimal tuning process selected $k = 1$, this configuration may increase sensitivity to local noise and reduce generalizability in real-world applications(Srivastava, 2024).Also, its inability to accommodate categorical variables could lead to the loss of information.

GBM

Gradient Boosting Machine (GBM) combines weak learners and minimizes prediction errors. This method iteratively corrects the mistakes of earlier trees using gradient descent. GBM can handle complex, non-linear relationships and achieve high accuracy. However, it can be computationally intensive and prone to overfitting without proper regularization. Its flexibility and predictive power make it a strong tool for high-dimensional features. We also implemented this method on our data; the same with the above analysis, we validated using repeated 5-fold cross-validation with 300 iterations. RFE method that was the same as kNN section was utilized to determine the best set of variables for GBM to use. The following variables were selected by RFE for the building of kNN classifier: FunctionalAssessment, ADL, MMSE, DietQuality, SleepQuality. We decided to add CholesterolHDL as an additional variable due to its association with AD. Due to GBM's nature as a tree-based algorithm, we implemented the a grid search approach (10-fold CV included), combined with a list of shrinkage parameter λ from 0.01 to 0.05, a list of number of trees from 1000 to 3000, to determine the best combination of shrinkage parameter and number of trees for the GBM classifier to use. The ideal combination was determined as

follows: $\lambda=0.05$, number of trees=1000. The repeated 5-fold Cross Validation with 300 iterations was utilized to validate the performance of GBM classifier.

Table 4. Performance Metrics of GBM classifier

Metrics	GBM
Recall	0.917
Recall CI	(0.9012,0.9023)
Precision	0.9285
Precision CI	(0.9279, 0.9291)
Accuracy	0.9406
Accuracy CI	(0.9404, 0.9409)
AUC	0.9495
AUC CI	(0.9493, 0.9497)

The GBM classifier achieved a recall of 0.9017 (95% CI: 0.9012,0.9023), a precision of 0.9285 (95% CI: 0.9279, 0.9291), an accuracy of 0.9406 (95% CI: 0.9404, 0.9409), and an AUC of 0.9495 (95% CI: 0.9493, 0.9497). Hence, this classifier was able to detect 90.17% of the positive AD cases in all patients with positive AD diagnosis status, 92.85% of the predicted positive cases were true positives. The accuracy of 0.9406 suggested that the classifier was able to correctly classify 94.06% of the patients into correct classes. The AUC of 0. 9495 indicated that the GBM demonstrated the strongest observed performance across all evaluated classifiers, suggesting that ensemble-based non-linear approaches may better capture complex predictor interactions within this dataset. However, its model complexity and computational demands may limit practical implementation without further validation and optimization.

Section 4. Conclusion

Five classifiers were constructed using MLR, MLR with best subset selection technique, MLR with Lasso regularization technique, kNN and GBM and validated their performance on classifying positive and negative cases of Alzheimer's disease. The performance of each classifier is displayed below.

Table 5. Summary of Performances of Classifiers

Metrics_Models	MLR	MLR BestSubset	Lasso	KNN	GBM
Recall	0.728	0.742	0.7286	0.637	0.917
Recall CI	(0.7273,0.7285)	(0.7416,0.7424)	(0.7281,0.7290)	(0.6368,0.6387)	(0.9169,0.9174)
Precision	0.793	0.8044	0.8083	0.642	0.947
Precision CI	(0.7919, 0.7931)	(0.8040, 0.8047)	(0.8078, 0.8088)	(0.6411, 0.6429)	(0.9464, 0.9469)
Accuracy	0.836	0.8446	0.8426	0.7458	0.9524
Accuracy CI	(0.8357, 0.8364)	(0.8440, 0.8447)	(0.8423, 0.8428)	(0.7452, 0.7463)	(0.9523, 0.9526)
AUC	0.8979	0.903	0.9011	0.721	0.9515
AUC CI	(0.8977, 0.8980)	(0.90288, 0.90304)	(0.90105,0.90125)	(0.7207, 0.7218)	(0.9513, 0.9516)

We can see from Table 5 above that the GBM classifier outperformed other classifiers across every performance metric. This suggests that GBM's mechanism of sequentially building weak learner models, where each learner corrects mistakes made by the previous one, is a robust approach that can be leveraged to construct a classifier for screening Alzheimer's disease using health-related measures. However, the computationally expensive nature of GBM models could hinder real-world application, and additional non-linear methods are worth considering. In addition, the synthetic nature of the dataset used for classifier construction may lead to reduced performance when the classifier is applied to real-world data; therefore, incorporating more real-world Alzheimer's disease records should be considered for the development of a more accurate and versatile classifier.

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